Deven Desai

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SUMMARY

Software Engineering Master's student and Stanford Hackathon winner specializing in high concurrency systems, low latency data retrieval, and full stack web architectures.

PROJECTS

High Performance Search Engine | Low Latency Discovery Platform GitHub

- Analyzed and queried high volume live inventory and event datasets in real time by engineering a custom full stack discovery interface as a solo developer using Python and React.
- Decreased search query response latency by 35 percent by implementing optimized indexing algorithms and memory structures as measured by end to end benchmark traces.
- Increased processing throughput by 20 percent by tuning lower level data structures and batch execution routines to support high concurrency requests.

AgenticFlow | Concurrent Distributed Microservices Engine GitHub

- Automated database transformations and data migrations across cloud systems by architecting an independent microservices engine using Go, Python, and gRPC.
- Increased system execution throughput by 40 percent by designing concurrent execution loops and low overhead microservice communication protocols.
- Maintained strict data consistency and privacy across system execution boundaries by engineering an automated validation layer verified via continuous testing.

OmniTrack AI | Full Stack Real Time Data Processing Platform GitHub

- Processed unstructured streaming data and user metadata natively in cloud infrastructure by engineering a zero to one full stack application using Python and React.
- Achieved sub 100ms vector retrieval latency by independently optimizing Qdrant indexing algorithms for compute intensive, concurrent requests.
- Enhanced system stability and output reliability by 30 percent by building continuous execution monitoring and automated testing suites.

WORK EXPERIENCE

Applied AI Intern, *Parspec, Inc., USA* **May 2026 – Dec 2026**

- Improved product-search relevance over a distributor inventory catalog, raising **recall@3 from 50% to 90%** on a **5,000-query** labeled evaluation set, by replacing keyword-only search with **hybrid** lexical and **semantic retrieval** fused via **reciprocal rank fusion**, re-scored with a **cross-encoder re-ranker**.
- Eliminated manual review of several-hundred-page construction documents for distributor bids, extracting structured product records at **88%** precision across **CSI Divisions 26 and 22**, by building a **RAG** pipeline over written specifications and schedules that grounds every extracted field in its source excerpt and auto-enriches contractor **Quantity Take-Off** line items.
- Automated lighting fixture product substitutions across 1.2M+ indexed datasheets at 85 percent match precision by engineering a retrieval pipeline that resolves free text inputs to anchor products and scores candidates via multi attribute filtering and semantic search.

Full Stack Developer, *Terminal Trends, India* **May 2024 – Dec 2024**

- Developed and optimized RESTful APIs using **Node.js** and **Express** to integrate real-time EV charging station data from multiple vendors (e.g., Tesla, government APIs), reducing server response time by **40%** and improving data reliability across the platform.
- Built interactive route-planning features with **React** and **Redux**, enabling users to locate nearby EV stations, view waiting times, pricing per kWh, and other live metrics along their journey, enhancing user convenience and engagement.

TECHNICAL SKILLS

Languages: Python, Go, TypeScript, JavaScript, C++, Java, C#, SQL, HTML5 CSS3
Backend & Systems: Distributed Systems, Microservices Architecture, RESTful APIs, gRPC, High Concurrency Pipelines
Frontend & Web: React, Redux, TailwindCSS, State Management, User Experience Polish, Web Performance
Databases & Storage: PostgreSQL, SQL Server, NoSQL, Redis, Vector Databases (Qdrant), Relational Schema Design

EDUCATION

San Jose State University **San Jose, CA**
Master of Science in Software Engineering

ACHIEVEMENTS

- **Winner – Davis Wright Tremaine LLP Prize**, *Stanford Codex LLM x Law Hackathon (2025)*
  Led a 4-member team to build a **RAG-based legal QA engine** using Python, GPT-4, and vector-embedding retrieval with **cosine similarity**, enabling accurate extraction of contract clauses and delivering **90%+ correct answers**, outperforming 30+ teams.